FujiNet for the ColecoVision (and the ADAM's cartridge slot), Rev0: an RP2350A cartridge on the Z80's bus + ESP32-S3.

The cart serves the console's fetches in software (fujinet-firmware pico/coleco): core1 drives D0-D7 off the bare address, the 74LVC245A lets the byte onto the bus only while a cart select (/8000 /A000 /C000 /E000, ANDed to /CE) is low, and DIR turns it round for the bank-register writes.

cart-bus	the 30-finger edge, 330R into GP0-14 (A0-A14), the 74LVC245A between the edge's D0-D7 and GP15-22, the select pull-ups and AND pair (/CE on GP26 and the buffer's /OE), DIR on GP27, the console-5V sense on GP28, the activity LED, the bring-up pads
rp-core	the RP2350A's supplies and decoupling, core regulator, the W25Q16 QSPI flash, crystal, SWD, RUN / BOOTSEL / RESET, its USB out to the S3
fujinet	the ESP32-S3 (the FujiNet, the RP's USB host), microSD, WS2812 status light
usb	USB-C, the USBLC6 ESD array, the CP2102N bridge and esptool's auto-program pair
power	+5V = the TPS2116 mux (USB VBUS first, console 5 V second), the RP's LDO (+3V3_RP), the S3's buck (+3V3)

Rails (GND, CONS_5V, VBUS, +5V, +3V3, +3V3_RP, DVDD) join by their power symbols.

Firmware contract (tools/check_nets.py checks this netlist against it): fujinet-firmware pico/coleco coleco_cart.h (A0-A14 GPIO0-14, D0-D7 GPIO15-22, LED 25, /CE 26, DIR 27, PWR 28); this board adds GP23 / GP24 (UART1 TX) / GP29 bring-up pads. The S3: fujiversal-coleco.h (SD 38-41, UART0 43 / 44, WS2812 48, USB host 19 / 20) + RUN / BOOTSEL on IO4 / IO5 and SD card detect on IO42 (README: firmware changes).

power

File: power.kicad_sch

fujinet

USB_DP
USB_DM
RUN_CTL
BOOTSEL_CTL
S3_EN
S3_IO0
S3_TXD
S3_RXD

File: fujinet.kicad_sch

rp-core

USB_DP
USB_DM
RUN_CTL
BOOTSEL_CTL
S3_EN

File: rp-core.kicad_sch

usb

S3_RXD
S3_TXD
S3_IO0
S3_EN

File: usb.kicad_sch

cart-bus

File: cart-bus.kicad_sch

the console cartridge slot (the 30-finger edge) is on cart-bus

Edge after MAME exp.h / ADAM TRM J1; outline after the ThatOldNerd, Commodore-Bench and Joe Gra
Rev0 2026-11-04: RP2350A + W25Q16 + ESP32-S3; firmware pico/coleco coleco_cart.h pin map
CERN-OHL-W-2.0; RP2350 regulator corner after Raspberry Pi RP2350A minimal design (MIT)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file

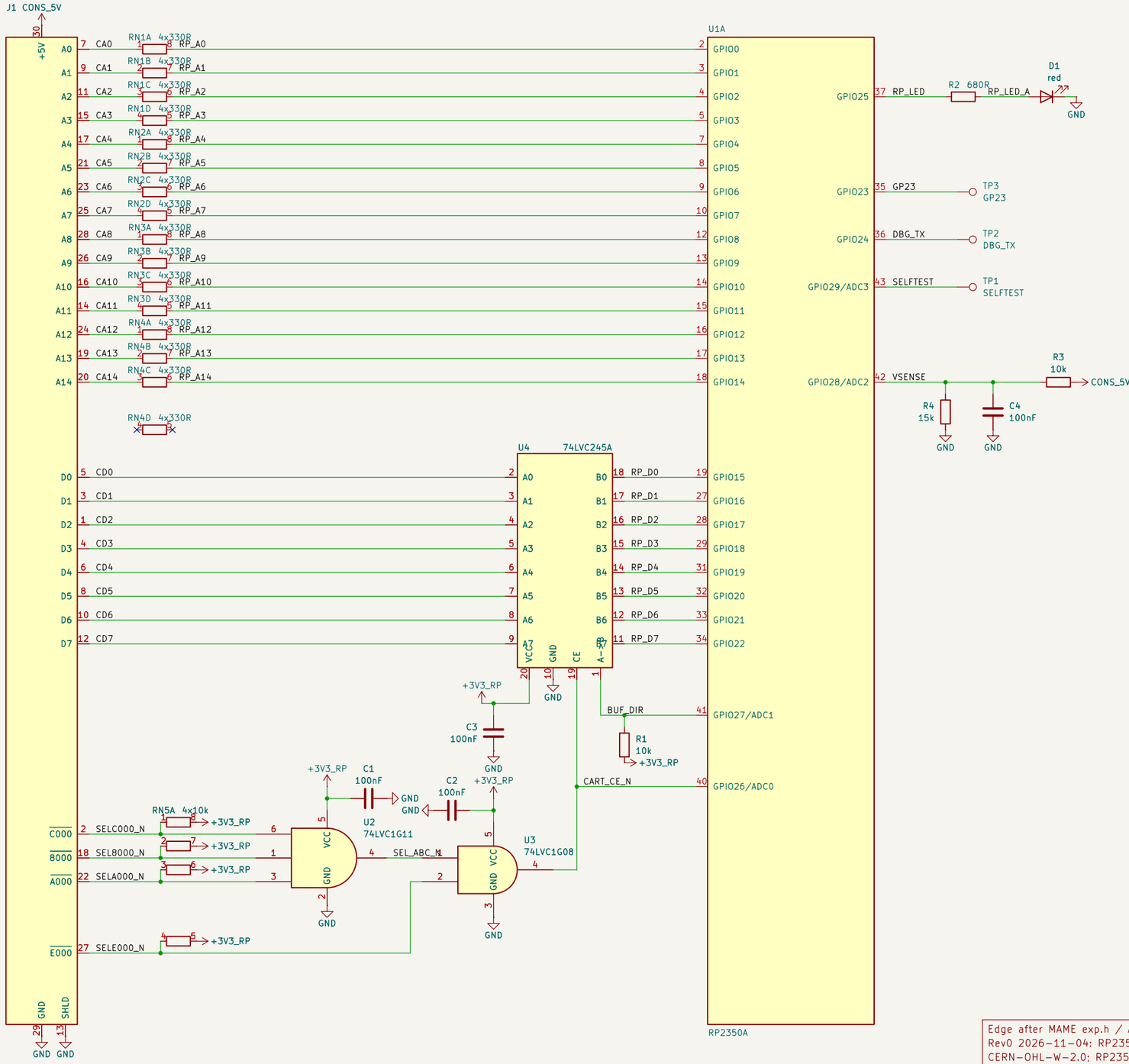
FujiNet

Sheet: /
File: FujiNet-ColecoVision-Rev0.kicad_sch

Title: FujiNet ColecoVision Rev0

Size: A3	Date: 2026-11-04	**Rev: 0**
KiCad E.D.A. 10.0.8		Id: 1/6

The ColecoVision cartridge port (MAME bus/coleco/cartridge/exp.h, ADAM Technical Reference Manual J1): even pins on the component face, odd pins behind them. A0–A14 and D0–D7 straight from the Z80; /8000 /A000 /C000 /E000 from the console's '138 (enabled by /MREQ and RFSH: a write selects as a read does). No /RD, /WR, A15, clock or reset. A0–A14 reach GP0–14 through 330R (5 V-tolerant pads). D0–D7 pass the 74LVC245A, whose /OE is the AND of the four selects: core1 drives GP15–22 off the bare address on every cycle (coleco_cart.c), and only a cart select lets the byte onto the bus. DIR (GP27, 1 = console -> cart, pulled up through reset) turns it round for the bank-register writes; /CE (GP26) tells core1 the cycle is the cart's. GP28 senses the console 5 V (10k / 15k = 3.0 V); GP25 the LED; GP23 / GP24 (UART1 TX) / GP29 bring-up pads.



Edge after MAME exp.h / ADAM TRM J1; outline after the ThatOldNerd, Commodore-Bench and Joe Gra
Rev0 2026-11-04: RP2350A + W25Q16 + ESP32-S3; firmware pico/coleco coleco_cart.h pin map
CERN-OHL-W-2.0; RP2350 regulator corner after Raspberry Pi RP2350A minimal design (MIT)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file

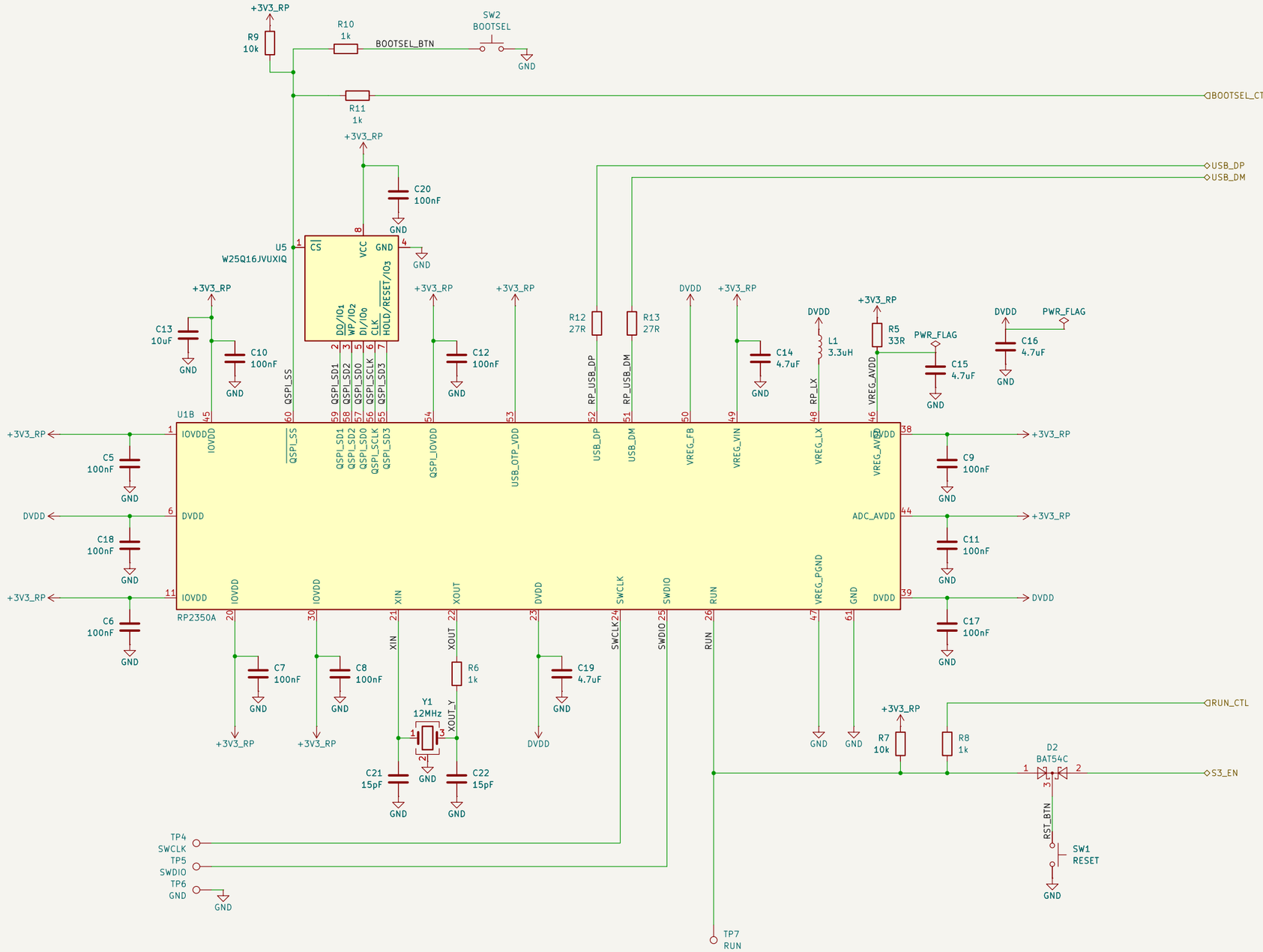
FujiNet

Sheet: /cart-bus/
File: cart-bus.kicad_sch

Title: FujiNet ColecoVision Rev0 – Cart bus: edge, select AND, data buffer, R

Size: A3	Date: 2026-11-04	**Rev: 0**
KiCad E.D.A. 10.0.8		Id: 2/6

Supplies (RP2350 datasheet 6.3.7): IOVDD, QSPI_IOVDD, USB_OTP_VDD, ADC_AVDD and VREG_VIN on +3V3_RP, the AP2112K LDO that rises with +5V; the core regulator switches VREG_LX through 3.3 uH onto DVDD (VREG_FB senses it), its analog supply VREG_AVDD through 33R / 4.7 uF. A capacitor at every supply pin (QSPI_IOVDD and USB_OTP_VDD share one, as in Raspberry Pi's minimal design), drawn at the pin it serves.
RESET resets the RP2350A and the ESP32-S3 together (D2 steers RST_BTN onto RUN and S3_EN): the way back to the FujiNet's CONFIG. Hold BOOTSEL while pressing RESET for the RP's USB boot ROM; the S3 does both itself: IO4 (RUN_CTL) low resets the RP, IO5 (BOOTSEL_CTL) low through that reset selects BOOTSEL. USB: 27R at the RP, the S3 is the host. SWD and RUN on bring-up pads. The W25Q16JV (2 MB, Pico 2-style boot2) on QSPI; its /CS is the strap.



Edge after MAME exp.h / ADAM TRM J1; outline after the ThatOldNerd, Commodore-Bench and Joe Gra
Rev0 2026-11-04: RP2350A + W25Q16 + ESP32-S3; firmware pico/coleco coleco_cart.h pin map
CERN-OHL-W-2.0; RP2350 regulator corner after Raspberry Pi RP2350A minimal design (MIT)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file

FujiNet

Sheet: /rp-core/
File: rp-core.kicad_sch

Title: FujiNet ColecoVision Rev0 — RP2350A core, QSPI flash, crystal, RUN, S

Size: A3	Date: 2026-11-04	**Rev: 0**
KiCad E.D.A. 10.0.8		Id: 3/6

ESP32-S3-WROOM-1-N16R8: the FujiNet (fujiversal-o2.h pin map), on +3V3 (the buck).
EN: 10k / 1 uF power-on delay. The S3_RST button, the cart RESET
(rp-core, through the BAT54C) and the USB bridge's auto-program pair
(usb sheet) all pull it low; S3_BOOT and the same pair pull IO0.
IO4 / IO5 drive the RP2350A's RUN and BOOTSEL (every on-cart S3's
pins); IO19 / IO20, the S3's own USB, host the RP's.
microSD in SPI mode, 10k pull-ups on CS, DO, DAT1, DAT2 and CD.
WS2812B-2020 on +5V: its data input (VIH 2.7 V) takes the S3's 3.3 V.
The 23 spare GPIOs (right) are not connected.

+3V3
R14
10k
S3_EN
C25
1uF
GND
+3V3
C24
100nF
C23
22uF
GND
GND
U6
EN
3V3
TXD0
RXD0
S3_TXD
S3_RXD
IO1
IO2
IO3
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO21
IO35
IO36
IO37
IO45
IO46
IO47
S3_IO0
IO0
SW3
S3_BOOT
GND
RUN_CTL
BOOTSEL_CTL
IO4
IO5
USB_DP
USB_DM
USB_D+
USB_D-
ESP32-S3-WROOM-1-N16R8
IO48
GND
IO42
IO41
IO40
IO39
IO38
LED_STRIP
GND
+3V3
RN6C
4x10k
+3V3
SD_DAT1
SD_DAT2
SD_MOSI
SD_SCK
SD_MISO
SD_CS
SD_CD
J2
DAT1
DAT2
CMD/DI
CLK
DAT0/DO
DAT3/CS
CD
VDD
VSS
microSD
+3V3
+3V3
+3V3
R15
10k
+5V
C27
100nF
GND
DOUT
VDD
DIN
VSS
WS_DIN
R16
330R
D3
WS2812B-2020-V6
GND
+3V3
C26
10uF
GND
GND

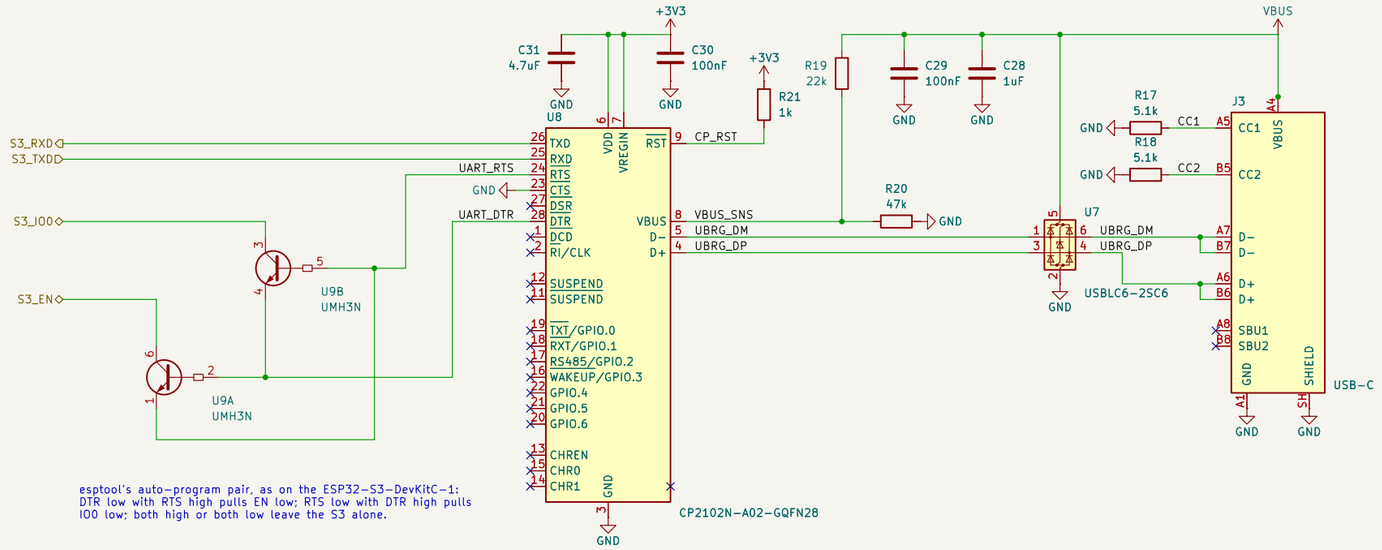
Edge after MAME exp.h / ADAM TRM J1; outline after the ThatOldNerd, Commodore-Bench and Joe Gra
Rev0 2026-11-04: RP2350A + W25Q16 + ESP32-S3; firmware pico/coleco coleco_cart.h pin map
CERN-OHL-W-2.0; RP2350 regulator corner after Raspberry Pi RP2350A minimal design (MIT)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file

FujiNet

Sheet: /fujinet/
File: fujinet.kicad_sch

Title: FujiNet ColecoVision Rev0 — FujiNet (ESP32-S3), microSD

Size: A3	Date: 2026-11-04	**Rev: 0**
KiCad E.D.A. 10.0.8		Id: 4/6



Edge after MAME exp.h / ADAM TRM J1; outline after the ThatOldNerd, Commodore-Bench and Joe Gra
Rev0 2026-11-04: RP2350A + W25Q16 + ESP32-S3; firmware pico/coleco coleco_cart.h pin map
CERN-OHL-W-2.0; RP2350 regulator corner after Raspberry Pi RP2350A minimal design (MIT)
Generated from tools/design.py + tools/sch_layout.py — edit those, not this file

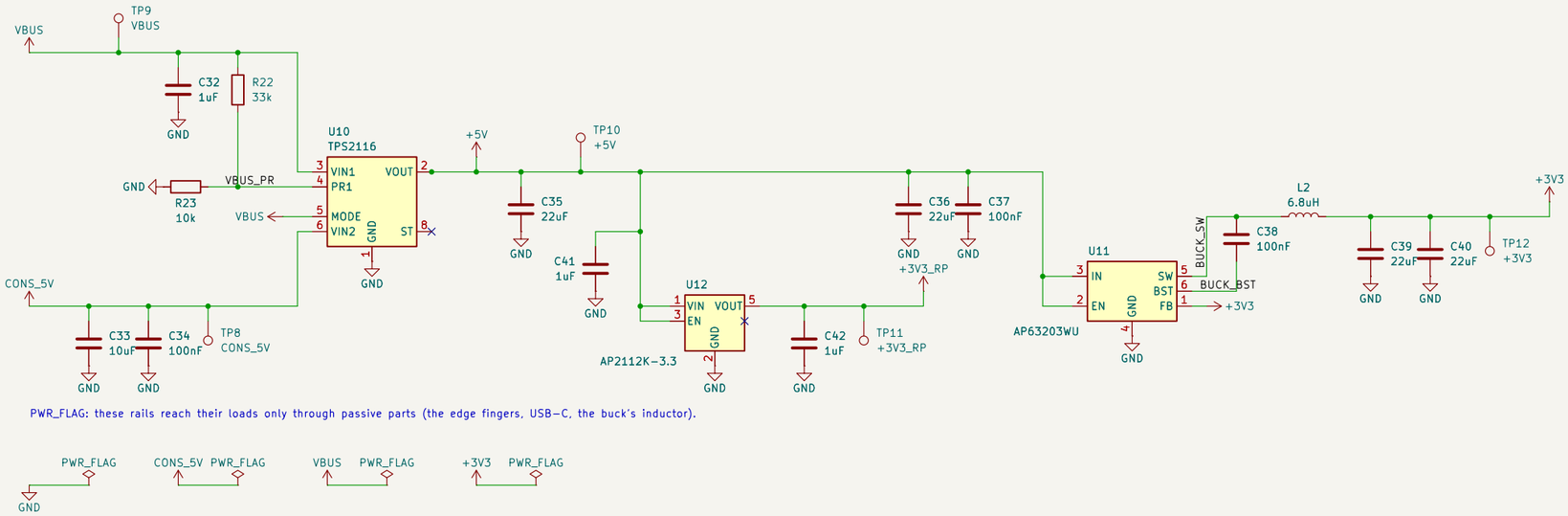
FujiNet

Sheet: /usb/
File: usb.kicad_sch

Title: FujiNet ColecoVision Rev0 — USB-C + CP2102N

Size: A3	Date: 2026-11-04	**Rev: 0**
KiCad E.D.A. 10.0.8		Id: 5/6

Power tree: +5V is the TPS2116 power mux's output (U10), USB first, the console's 5 V second.
  VBUS   USB-C (usb sheet) -> VIN1, the priority input: MODE on VIN1, PR1 from R22 33k / R23 10k, so VBUS
            above VREF x 4.3 (3.96-4.64 V) runs the cart and the console's supply delivers nothing.
  CONS_5V edge pin D, the console's 5 V (spare current not published: prefer USB) -> VIN2, without USB.
  37-60 mOhm either way: reverse current blocking: neither source back-feeds the other, so a cart on USB
  never powers a console that is off. ST is not used.
+5V feeds the WS2812, U12 AP2112K -> +3V3_RP (every RP2350A supply pin: it rises with +5V, so the 5 V-tolerant
pads are never unpowered with 5 V on them) and U11 AP63203 -> +3V3 (the ESP32-S3, the CP2102N and the microSD;
WiFi bursts ~400 mA; it waits out UVLO, then soft-starts over 4 ms). Without USB the cart adds ~145 mA average,
~370 mA peak to the console's supply (tools/audit/margins.py): prefer USB power. TP9, TP8, TP10, TP11 and TP12 are
bring-up pads.



PWR_FLAG: these rails reach their loads only through passive parts (the edge fingers, USB-C, the buck's inductor).

Edge after MAME exp.h / ADAM TRM J1; outline after the ThatOldNerd, Commodore-Bench and Joe Gra
Rev0 2026-11-04: RP2350A + W25Q16 + ESP32-S3; firmware pico/coleco coleco_cart.h pin map
CERN-OHL-W-2.0; RP2350 regulator corner after Raspberry Pi RP2350A minimal design (MIT)
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file
FujiNet

Sheet: /power/		
File: power.kicad_sch		
Title: FujiNet ColecoVision Rev0 - Power: mux, S3 buck, RP LDO		
Size: A3	Date: 2026-11-04	**Rev: 0**
KiCad E.D.A. 10.0.8		Id: 6/6